

ACIPAN INJECTION 1 ML

DESCRIPTION:

ACIPAN INJECTION 1 ML : A clear, colourless and odourless solution

COMPOSITION:

Each ml contains Atropine Sulphate 1 mg

PHARMACODYNAMICS:

Atropine is an antimuscarinic alkaloid. It competitively inhibits the actions of acetylcholine and has both peripheral and central actions. It reduces secretions especially salivary and bronchial secretions, and also reduces perspiration. It has little effect on biliary or pancreatic secretion.

Atropine has an antispasmodic action on smooth muscle and diminishes gastric and intestinal motility. It blocks the responses of the sphincter muscle of the iris and ciliary muscle of the lens to cholinergic stimulation thereby dilating the pupil (mydriasis) and paralysing accommodation (cycloplegia).

Atropine alters cardiac rate, initially slowing the heart rate. However, the dominant effect is tachycardia which occurs with average clinical doses (400-600 microgram) and is more profound with larger doses.

Atropine has central nervous system (CNS) activity. Average doses (400 microgram - 1 mg) produce stimulation of the medulla and higher cerebral centres, causing mild vagal excitation. Larger doses cause prominent central excitation, blocking the vagus nerve resulting in restlessness, irritability, disorientation, hallucinations or delirium.

PHARMACOKINETICS:

Absorption: Atropine is well absorbed following intramuscular administration. Peak plasma are reached within 30 minutes.

Inhibition of salivation (by I.M. administration): occurs – within 30 minutes. Peaks – within 1 – 2 hours. Duration – persists for 4 hours.

Increased heart rate: By I.M. administration: occurs – within 5 – 40 minutes. Peaks – within 20 minutes – 1 hour.

By I.V. administration: occurs – within 2 – 4 minutes

Distribution: Atropine is well distributed throughout body and crosses the blood-brain barrier and also the placenta barrier.

Metabolism: Atropine is metabolized in liver to several metabolites.

Excretion: Plasma half life – 2 – 3 hours. Atropine appears biphasic with an initial phase of ~ 2 hours and half life in the terminal phase of 12.5 hours or longer.

Mainly excreted in urine (~ 30 – 50% of a dose excreted unchanged).

INDICATIONS:

1. Pre-anaesthetic medication to inhibit excessive salivary and bronchial secretion and to diminish the risk of vagal inhibition of the heart.
2. Management of patients with acute myocardial infarction and sinus bradycardia who have associated hypotension and increased ventricular irritability.
3. In conjunction with morphine or other drugs for the relief of biliary or renal colic.
4. Administered concurrently with anticholinesterase agents (eg. neostigmine, physostigmine) to block the adverse muscarinic effects of these latter agents when they are used after surgery to terminate curarization.
5. Concomitant with a cholinesterase reactivator (eg.: pralidoxime) to reverse muscarinic effects associated with toxic exposure to anticholinesterase compounds (eg.: organophosphate pesticides).

RECOMMENDED DOSAGE:

Atropine Sulphate Injection may be given by subcutaneous, intramuscular or direct intravenous injection. The intravenous injection should be administered slowly. Atropine Sulphate Injection should not be added to any intravenous infusion solutions for administration.

Cardiopulmonary Resuscitation: The usual adult dose is 0.5-1 mg IV, which may be repeated at 5 minutes intervals until the desired heart rate is achieved. The total dose should not exceed 2 mg. The usual pediatric dose is 0.02 mg/kg (maximum 0.5 mg) IV, which may be repeated at 5 minute intervals until the desired heart rate is achieved. The total dose should not exceed 1 mg.

Premedication: 300 to 600 micrograms Atropine Sulfate Injection may be given intramuscularly or subcutaneously 30 to 60 minutes prior to induction of anesthesia. Alternatively 300 to 600 micrograms intravenously may be given immediately before induction of anesthesia.

Children: 65 micrograms atropine subcutaneously for premature infants, 100 micrograms for full term infants and 200 micrograms for infants between 6 months to 1 year. In children older than 1 year atropine may be given intramuscularly or subcutaneously at 10 to 20 micrograms/kg.

Reversal of Competitive Neuromuscular Block: May be given by slow intravenous injection in conjunction with an anticholinesterase agent (eg. neostigmine, physostigmine). 600 micrograms to 1.2 mg atropine for each 0.5 to 2.5 mg neostigmine methylsulfate in adults and 0.02 mg/kg atropine for each 0.04 mg/kg neostigmine methylsulfate in children.

Organophosphate Poisoning: 1 to 2 mg atropine may be given intravenously. Additional 2 mg doses may be administered intramuscularly or intravenously every 5-60 minutes until symptoms subside; and repeated if they reappear. For severe cases 2-6 mg may be administered intravenously, with subsequent additional doses of 2-6 mg being administered until symptoms subside.

Doses up to 50 mg may be required within the first 24 hours. With severe cases atropine therapy should be withdrawn gradually to avoid sudden recurrence of symptoms. A cholinesterase reactivator (eg. pralidoxime) is administered concomitantly.

The dose for children is 0.05 mg/kg IM or IV, repeated at 10 to 30 minute intervals until symptoms subside.

ROUTE OF ADMINISTRATION : For s.c, i.m. or slow i.v. injection

CONTRAINDICATIONS:

- i. Known sensitivity to atropine or other anticholinergic agents.
- ii. Severe ulcerative colitis.
- iii. Toxic megacolon complicating ulcerative colitis.
- iv. Gastrointestinal obstruction eg pyloroduodenal stenosis, achalasia, cardiospasm, paralytic ileus, intestinal atony.
- v. Closed-angle glaucoma
- vi. Obstructive uropathy eg bladder neck obstruction caused by prostatic hypertrophy.
- vii. Myasthenia gravis
- viii. Tachycardia secondary to cardiac insufficiency or thyrotoxicosis.
- ix. Acute hemorrhage with unstable cardiovascular status.

WARNINGS AND PRECAUTIONS:

Atropine should be used with caution in geriatric patients and children since they may be more susceptible to its adverse effects. Patients with elevated ambient temperature should not receive atropine due to the risk of provoking hyperpyrexia, especially in children. Atropine should be used with caution in conditions characterised by tachycardia and in cardiac surgery where it may further accelerate the heart rate. Tachycardia may result from vagal inhibition and induce angina pectoris in patients with coronary heart disease. Atropine may cause mental confusion, especially in elderly or brain damaged patients. Caution is required in administering atropine to debilitated patients with chronic pulmonary disease, since reduced bronchial secretions may cause inspissation and bronchial plug formation. Where patients are exposed to high environmental temperatures, atropine should be used cautiously since heat prostration can result. Atropine should be used with caution in patients with hyperthyroidism, hepatic or renal disease, hypertension or autonomic neuropathy. Since atropine decreases gastrointestinal motility, it should be used with caution in patients with gastric ulcer, oesophageal reflux, known or suspected gastrointestinal infections, incomplete intestinal obstruction or ulcerative colitis. Atropine should also be used with caution in patients with diarrhoea, since diarrhoea may be an early symptom of incomplete intestinal obstruction, especially in patients with ileostomy or colostomy. Since antimuscarinics may cause drowsiness or blurred vision, patients should be warned not to engage in activities requiring mental alertness and/or visual acuity (eg driving a car or operating machinery) while taking drugs. Systemic administration of conventional doses of atropine may precipitate acute glaucoma in susceptible individuals.

INTERACTIONS WITH OTHERS MEDICAMENTS:

Atropine may cause increased anticholinergic activity when administered concomitantly with other anticholinergic drugs such as phenothiazines, antiparkinsonian drugs, antiarrhythmics with anticholinergic activity eg disopyramide and quinidine, antihistamines, tricyclic antidepressants or butyrophenones. The absorption of other drugs may be affected by the reduction in gastric motility caused by atropine. Atropine antagonises the actions of a number of compounds, including: synthetic choline esters eg bethanechol and carbachol, anticholinesterase drugs eg physostigmine, neostigmine and pyridostigmine, and cholinomimetic alkaloids eg pilocarpine.

USE IN PREGNANCY AND LACTATION:

Use in Pregnancy: Atropine has been used in a large number of pregnant women and women of child-bearing age without an increase in the frequency of malformations or other direct or indirect harmful effects on the foetus having been observed. However, proven safety of atropine in pregnancy has not been established. As with all drugs caution is essential in the use of atropine in pregnant women.

Use in Lactation: Atropine is reported to distribute into breast milk in small quantities. Since some infants can be sensitive to atropine, breastfeeding cannot be recommended. Atropine may cause antimuscarinic effects in the infant and inhibit lactation.

SIDE EFFECTS:

More Common Reactions: Xerostomia, blurred vision, photophobia, cycloplegia, mydriasis, urinary hesitancy and retention, flushing, tachycardia, palpitations and constipation. These effects may appear at therapeutic or subtherapeutic doses.

Less Common Reactions: Tremor, headache, nervousness, drowsiness, weakness, hyperpyrexia, dizziness, confusion and/or excitement, increased ocular tension, loss of taste, nausea. Anaphylaxis has been reported. Urticarial rash which may progress to exfoliation, and various dermal manifestations have also been reported.

Atropine blocks vagal impulses with consequent increase in heart rate.

SYMPTOMS AND TREATMENT OF OVERDOSE:

There is considerable patient variability in the susceptibility to atropine. Atropine overdosage is characterised by both peripheral and central symptoms. Toxic doses cause tachycardia, rapid respiration, hyperpyrexia, and central nervous system stimulation marked by restlessness, confusion excitement, paranoid psychotic reactions, delirium, hallucinations and occasionally seizures or convulsions. A rash may appear on the face or upper trunk. In severe toxicity, CNS stimulation may give way to CNS depression, coma, circulatory and respiratory failure and death. Treatment: Symptomatic and supportive therapy should be provided. Close monitoring, including ECG monitoring is recommended. Fluid therapy and other standard treatments for shock should be administered. Hyperthermia should be treated with cold packs, mechanical cooling devices or sponging with tepid water. Maintenance of adequate airway is essential, with respiratory assistance if necessary. Urinary catheterisation may be required if the patient is comatose. The use of physostigmine as an antidote for atropine poisoning is controversial due to the potential for physostigmine to produce severe adverse effects. The use of physostigmine should be reserved for treatment of patients with extreme delirium or agitation, patients with repetitive seizures, patients with severe sinus tachycardia or supraventricular tachycardia or unresponsive extreme hyperthermia. Relative contra-indications to the use of physostigmine include asthma, gangrene, cardiovascular disease and mechanical obstruction of the gastrointestinal or genito-urinary tract. In such patients physostigmine should only be used where a life threatening emergency occurs. Diazepam may be administered to control excitement, delirium or other symptoms of acute psychosis. Phenothiazines should be avoided. Delirium, hallucinations and cardiac arrhythmias often respond to physostigmine. Physostigmine will reverse supraventricular tachycardia produced by atropine overdosage. Intravenous propranolol may be useful for treatment of supraventricular tachyarrhythmias unresponsive to physostigmine or where physostigmine is contraindicated.

INCOMPATIBILITIES:

Atropine sulfate injection has been shown to be incompatible with solutions containing adrenaline hydrochloride, amylbarbitone sodium, ampicillin sodium, chloramphenicol sodium succinate, chlortetracycline hydrochloride, heparin sodium, metaraminol tartrate, methicillin sodium, nitrofurantoin, novobiocin, oxacillin, sodium bicarbonate, sulfadiazine sodium, sulfafurazole diethanolamine, tetracycline hydrochloride, thiopentone sodium, vitamin B complex with ascorbic acid, warfarin sodium.

STORAGE CONDITIONS:

Store below 30°C. Protect from light. Avoid freezing.

Keep out of reach of children. *Jauhkan daripada kanak-kanak.*

PACK SIZES:

Pack in 10 ampoules of 1 ml / box.

SHELF LIFE:

Please refer to outer package.

PRODUCT REGISTRATION HOLDER AND MANUFACTURER:

DUOPHARMA (M) SDN BHD
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